

Cloud Platform Architecture over Virtualized Data Centers

Module 3.1 — Comprehensive Study Notes

Reference: Distributed and Cloud Computing — Kai Hwang, Geoffrey C. Fox, Jack Dongarra

1. Data-Center Design — Warehouse-Scale Data Centers

A data center is often built with a large number of servers through a huge interconnection network. The study covers large-scale warehouse data centers and small modular data centers that can be housed in a 40-ft truck container.

1.1 Overview of Warehouse-Scale Data-Center Design

Dennis Gannon: "The cloud is built on massive data centers."

- A warehouse-scale data center (WSC) is as large as a shopping mall — approximately 11 times the size of a football field, all under one roof.
- Can house between 400,000 to 1 million servers.
- Built on the principle of economics of scale — lower unit cost for larger data centers.
- A small data center may house as few as 1,000 servers for comparison.

Key Point: *The larger the data center, the lower the operational cost per server.*

1.2 Economics of Scale

Monthly operational cost estimates for a large data center (400,000+ server scale):

- Network cost: \$13/Mbps
- Storage cost: \$0.4/GB
- Plus administration/operational costs

Comparing large vs. small (1,000-server) data centers:

- Network cost of a small data center is approximately 7x greater per unit.
- Storage cost of a small data center is approximately 5.7x greater per unit.

Key Point: *Inference: The larger the data center, the lower the operational cost. This justifies investment in warehouse-scale facilities by cloud providers.*

1.3 Data-Center Construction Requirements

Most data centers are built with commercially available (off-the-shelf) components:

- Off-the-shelf server consists of: processor sockets with multicore CPUs, cache hierarchies, local shared DRAM, and directly attached disk drives.
- DRAM and disk resources within a rack are accessible through first-level rack switches.
- All resources in all racks are accessible via a cluster-level switch.

Example configuration — 2,000 server data center:

- Each server: 8 GB DRAM and four 1 TB disk drives

- 40 servers per rack, connected via 1 Gbps link to a rack-level switch
- Each rack switch has an additional 8×1 Gbps ports to connect the rack to the cluster-level switch

1.4 Cooling System of a Data-Center Room

Cooling is one of the most critical and expensive aspects of data center operation.

Raised Floor Design

- Data-center rooms use raised floors to hide cables, power lines, and cooling supplies.
- The raised floor is a steel grid resting on stanchions about 2–4 ft above the concrete floor.
- Under-floor area primarily distributes cool air to server racks; also routes power cables.

CRAC Unit (Computer Room Air Conditioning)

- CRAC units pressurize the raised floor plenum by blowing cold air into it.
- Cold air escapes the plenum through perforated tiles placed in front of server racks.
- Hot air from servers circulates back to the CRAC unit intakes, gets cooled, and is exhausted back into the plenum.
- Incoming coolant temperature: 12–14°C; warm coolant returns to a chiller.

Hot Aisle / Cold Aisle Arrangement

- Racks are arranged in long aisles alternating between cold aisles and hot aisles.
- This arrangement avoids mixing hot and cold air, improving cooling efficiency.
- Cold aisles face the perforated tiles (cool air intake side).
- Hot aisles face the rear of the servers (exhaust side) and direct heat back to CRAC units.

Key Point: *Proper hot/cold aisle arrangement is essential to avoid "hot spots" and wasted cooling energy.*

2. Data-Center Interconnection Networks

A critical core design of a data center is the interconnection network among all servers in the cluster.

2.1 Five Core Requirements for Interconnection Networks

- **Low Latency:** Minimal delay in data transmission between nodes.
- **High Bandwidth:** Large capacity to transfer data rapidly across the network.
- **Low Cost:** Network infrastructure should be economical to deploy and maintain.
- **MPI Communication Support:** Must support Message Passing Interface (MPI) patterns used in distributed computing.
- **Fault Tolerance:** Network must continue to function even when components fail.

2.2 Specific Design Considerations

(a) Application Traffic Support

- The topology must support all MPI communication patterns: both point-to-point and collective communications.
- The network should have high bisection bandwidth to meet traffic demands.

- One-to-many communications support distributed file access; one or a few servers act as metadata master servers and communicate with slave server nodes.
- To support MapReduce: network must perform map and reduce functions at high speed.
- The underlying network must support various traffic patterns demanded by different cloud applications.

(b) Network Expandability

- The interconnection network should be expandable as more servers are added.
- With thousands or hundreds of thousands of nodes, the network must be restructured to accommodate growth.
- Fat-tree and crossbar networks can be implemented with low-cost Ethernet switches.
- However, expansion becomes challenging when the number of servers increases sharply.

(c) Fault Tolerance

- The network must tolerate link or switch failures via multiple paths between any two server nodes.
- Fault tolerance of servers achieved by replicating data and computing among redundant servers.
- Both software and hardware redundancy must apply.
- The software layer must be aware of network failures; packet forwarding avoids broken links.
- Software drivers handle failures transparently without affecting cloud operations.

(d) Graceful Degradation

- The network should degrade gracefully amid limited node failures (not collapse entirely).
- Hot-swappable components are desired (components can be replaced without shutting down the system).
- No critical paths or single points of failure that can pull down the entire system.

(e) Switch-centric vs. Server-centric Design

Switch-Centric Design:

- Switches are used to connect server nodes.
- No modifications needed to the servers or their operating systems.
- Cleaner separation between network and compute layers.

Server-Centric Design:

- Modifies the operating system running on the servers.
- Special drivers are designed for relaying traffic through servers.
- Switches still need to be organized, but servers act as network participants too.

Key Point: *BCube is an example of a server-centric design used in modular data centers.*

3. Fat-Tree Interconnection Topology for Warehouse-Scale Data Centers

The fat-tree topology is the primary interconnection design applied to large-scale warehouse data centers.

3.1 Fat-Tree Topology Structure

- Organized into two main layers: server nodes layer (bottom) and switch layers (top).
- Three types of switches:
 - Edge Switches: Connect server nodes at the bottom layer.
 - Aggregation Switches: Upper layer that aggregates the lower-layer edge switches.
 - Core Switches: Provide paths between different pods.
- A pod consists of a group of aggregation switches, edge switches, and their leaf (server) nodes.

3.2 Fault Tolerance in Fat-Tree

- The fat-tree structure provides multiple paths between any two server nodes.
- Multiple paths provide fault-tolerant capability — alternate path is used if a link fails.
- Failure of an aggregation switch does NOT affect connectivity of the whole network.
- Failure of an edge switch only affects a small number of end server nodes.
- Extra switches in a pod provide higher bandwidth for cloud applications.

Key Point: *Fat-tree is the go-to topology for warehouse-scale data centers due to its redundancy, scalability with low-cost Ethernet switches, and multiple-path fault tolerance.*

4. Modular Data Centers (MDC) in Shipping Containers

4.1 Overview

- A modern data center can be structured as a shipyard of server clusters housed in truck-towed containers.
- SGI ICE Cube is a well-known example of modular data center design.
- A modular data center (MDC) can be placed anywhere data center capacity is needed.
- Available in 40-foot portable and multi-story configurations.

4.2 Internal Structure of the Container

- Inside the container, hundreds of blade servers are housed in racks surrounding the container walls.
- An array of fans forces heated air from server racks through a heat exchanger.
- The heat exchanger cools the air, which circulates back to the next rack in a continuous loop.

Rack Server vs. Blade Server:

- **Rack Server:** Built to be mounted within a server rack; general-purpose, configured for wide range of requirements; secured with mounting screws or rails.
- **Blade Server:** Modular server housed in a chassis; physically thin with only CPUs, memory, integrated network controllers, and sometimes storage; video cards/other components facilitated by the chassis.

4.3 Capacity

- The SGI ICE Cube container can house 46,080 processing cores OR 30 PB of storage per container.
- Large-scale data centers built with multiple modular containers resemble a big shipping yard.

4.4 Motivation for Container-Based Data Centers

- Demand for lower power consumption.
- Higher computer density.
- Mobility: relocate data centers to better locations with lower electricity costs.
- Better cooling water supplies.
- Cheaper housing for maintenance engineers.
- Data centers can be built at sites where leases and utilities are cheaper and cooling is more efficient.
- High degree of mobility — not fixed at a location unlike traditional data centers.

4.5 Container Data-Center Construction

- Container is truck-towable; designed to be weatherproof and easy to transport.
- The modular container design integrates network, computer, storage, and cooling gear.
- Construction is staged:
 - Start with a single server system
 - Move to rack system design (40 servers per rack — takes about half a day to build)
 - Finally scale to a full container system with multiple racks for ~1000 servers
- Full container system requires layout of floor space for power, networking, cooling, and complete testing.
- MDC construction and testing can be completed in a few days with all components, power, and water available.
- Cooling: Increase efficiency by varying water and airflow; must meet seasonal load requirements.

4.6 Portability and Applications

- The MDC approach supports many cloud service applications.
- Example: The healthcare industry benefits from installing data centers at all clinic sites.
- Edge computing deployments where proximity to end users is required.

5. Comparison: Modular Data Center vs. Warehouse-Scale Data Center

Aspect	Modular Data Center (MDC)	Warehouse-Scale Data Center (WSC)
Definition	Composed of standardized, pre-engineered modules/containers integrating servers, cooling, power & networking	Large centralized computing facility housing tens/hundreds of thousands of servers; operates as one massive system

Design	Container-based or prefabricated units; added/removed as demand grows	Designed as a large warehouse-like facility; entire infrastructure planned and deployed at large scale
Scalability	Incremental scaling - capacity expanded by adding modules	Massive scaling - designed to support extremely large workloads from the beginning
Deployment	Faster - modules are pre-built and only require installation and connection to utilities	Longer - infrastructure must be constructed and configured as a large facility
Typical Usage	Rapid deployment, edge computing, or organizations needing flexible expansion	Large cloud service providers (Google, Amazon, Microsoft) for large-scale internet services
Management	Each module is a self-contained unit with its own power, cooling, and networking	Entire data center treated as one large computer; thousands of servers cooperate for distributed applications

6. Server-Centric BCube Network for Modular Data Centers

6.1 BCube Architecture

- BCube is a high-performance, server-centric network specifically designed for modular data centers.
- Reference: C. Guo et al., ACM SIGCOMM Computer Communication Review, Oct. 2009.
- In BCube diagrams: servers are represented by circles, switches by rectangles.
- BCube provides a layered (hierarchical) structure.

BCube Layered Structure:

- The bottom layer contains all server nodes — forms Level 0.
- Level 1 switches form the top layer of BCube0.
- Higher-level BCubes (BCube1, BCube2...) are constructed by connecting multiple BCube0 units.

6.2 Key Features of BCube

- **Multiple Paths:** BCube provides multiple paths between any two nodes, offering extra bandwidth and fault tolerance.
- **OS Kernel Module:** BCube provides a kernel module in the server OS to perform routing operations — hence server-centric.
- **Cloud Application Support:** Multiple paths provide extra bandwidth to support various communication patterns in cloud applications.
- **Building Block Design:** BCube is designed as a building block for modular data centers, used inside server containers.

6.3 Inter-Module Connection Networks — MDCube

- BCube is used inside individual server containers.
- Containers themselves are the building blocks for larger data centers.

- A higher-level interconnection is needed to connect multiple containers together.
- MDCube (Modularized Datacenter Cube) was proposed for inter-container networking.
- MDCube connects multiple BCube containers using high-speed switches within the BCube.
- Example: A 2-D MDCube can be constructed from 9 BCube containers arranged in a grid.

Key Point: *BCube handles intra-container networking; MDCube handles inter-container networking. Together they form a complete network architecture for large-scale modular data centers.*

7. Data-Center Management Issues

Basic requirements for managing data center resources:

- **1. Making Common Users Happy:** The data center should be designed to provide quality service to the majority of users for at least 30 years (long-term service reliability).
- **2. Controlled Information Flow:** Information flow should be streamlined. Sustained services and High Availability (HA) are the primary goals.
- **3. Multiuser Manageability:** The system must be managed to support all data center functions: traffic flow management, database updating, and server maintenance — simultaneously for multiple users.
- **4. Scalability for Database Growth:** The system must allow growth as workload increases. Storage, processing, I/O, power, and cooling subsystems should all be independently scalable.
- **5. Reliability in Virtualized Infrastructure:** Failover, fault tolerance, and VM live migration should be integrated to enable recovery of critical applications from failures or disasters.
- **6. Low Cost to Users and Providers:** All operational costs of the cloud system built over data centers should be minimized, including hardware, power, networking, and administration costs.
- **7. Security Enforcement and Data Protection:** Data privacy and security defense mechanisms must be deployed to protect against network attacks and system interrupts, and to maintain data integrity from user abuses or network attacks.
- **8. Green Information Technology:** Saving power consumption and upgrading energy efficiency are in high demand when designing and operating current and future data centers. This includes renewable energy, optimized cooling, and energy-proportional computing.

Key Point: *Management issues represent the operational challenges that arise after a data center is built. Both technical (fault tolerance, VM migration) and non-technical (cost, green IT) concerns must be addressed.*

8. Quick Revision — Key Terms and Definitions

- **WSC (Warehouse-Scale Data Center):** A large centralized computing facility housing 400K–1M servers, operating as a single massive computing system. Used by Google, Amazon, Microsoft.
- **MDC (Modular Data Center):** A portable, container-based data center using pre-engineered modules. Can be deployed anywhere and scaled incrementally.
- **Economics of Scale:** Larger data centers have lower per-unit operational cost; network cost 7x lower and storage cost 5.7x lower compared to small centers.
- **CRAC Unit:** Computer Room Air Conditioning unit that pressurizes the raised floor plenum with cold air to cool server racks.
- **Hot/Cold Aisle:** Server rack arrangement alternating cold air intake aisles and hot air exhaust aisles to prevent mixing and improve cooling efficiency.
- **Fat-Tree Topology:** A switch-centric hierarchical topology (edge → aggregation → core switches + pods) providing multiple paths, fault tolerance, and high bandwidth. Used in WSC.
- **BCube:** A server-centric, layered network architecture for modular data centers; routing handled by a kernel module in server OS; provides multiple paths and high bandwidth.
- **MDCube:** Inter-container network topology connecting multiple BCube containers using high-speed switches; a 2-D MDCube uses 9 BCube containers.
- **MPI:** Message Passing Interface — communication standard for parallel computing; data center networks must support both point-to-point and collective MPI patterns.
- **Bisection Bandwidth:** The bandwidth across the middle split of a network; high bisection bandwidth is required to support all-to-all communications.
- **Graceful Degradation:** The ability of a network/system to continue operating (at reduced capacity) when components fail, without total system collapse.
- **Hot-Swappable:** Components that can be replaced/upgraded without shutting down the system, enabling continuous availability.
- **VM Live Migration:** Moving a running virtual machine from one physical server to another without downtime, critical for fault tolerance and load balancing.
- **High Availability (HA):** System design goal ensuring services remain accessible and operational with minimal downtime.
- **Blade Server:** A thin, modular server card that slides into a shared chassis; CPUs and RAM integrated on blade; power and I/O shared from chassis.
- **Rack Server:** A standard server mounted in a rack; general-purpose and configurable; secured with screws or rails.